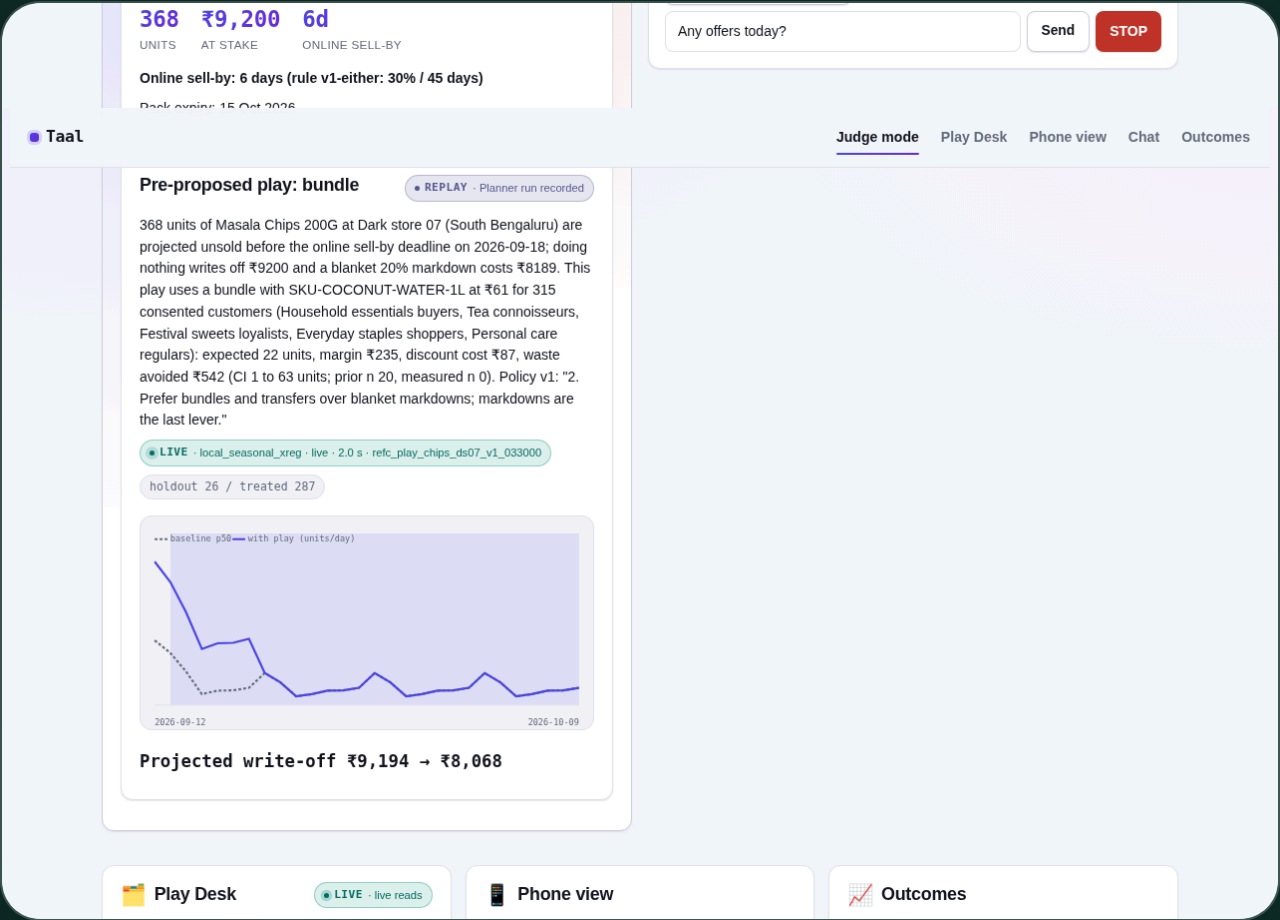


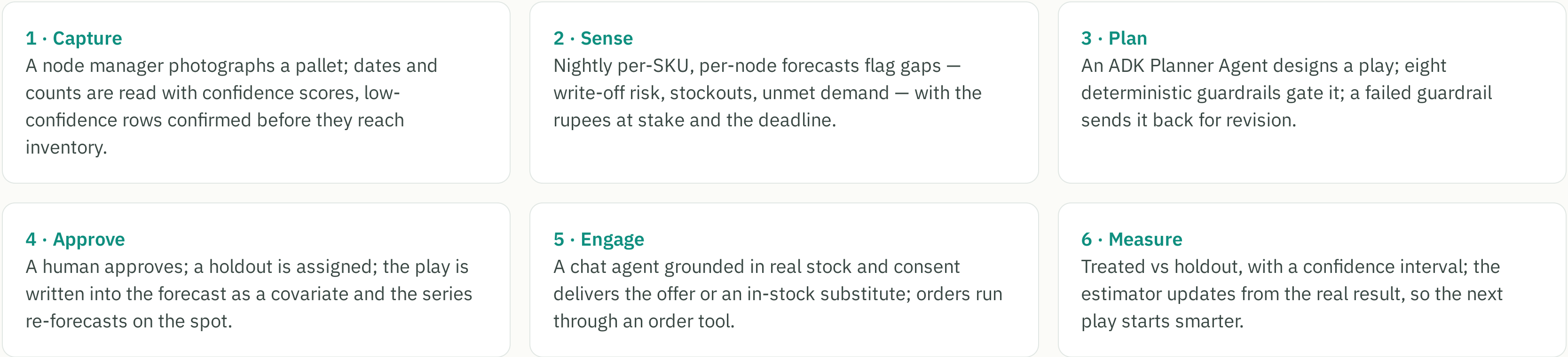
Taal

The agent that sells what the forecast says you'll throw away.



WHAT IT DOES

Six steps from a forecast to a sale



ONE REAL RUN, THIS TENANT ₹9,194 → ₹8,068 projected write-off on one lot, before and after a live guardrail-approved play

THE HOOK

33 days on the pack.
6 days to sell it online.

India's food regulator requires 30% of shelf life or 45 days remaining at delivery for food sold online. No forecasting tool knows that deadline exists — Taal computes it per lot, versioned as a tenant policy, and shows it on every gap card.

Demo lot, dark store DS-07

368 units

Masala Chips · ₹9,200 at stake

INNOVATION

What makes this a decision engine, not a promo bot

A legal deadline as a model input

The FSSAI online sell-by rule is computed per lot and, once a play is approved, written straight into the forecast as a covariate — the series re-forecasts on the spot, not on the next nightly run.

A holdout on every play

No play passes the gate without a randomized holdout. Lift is treated minus holdout with a confidence interval, and when the data says null the Outcomes screen reports a null — not an assumed lift.

An estimator that learns

Every Measure run updates the estimator's priors from real treated-vs-holdout counts, so the next play of that shape starts from evidence, not a static assumption.

Not a fourth marketing channel

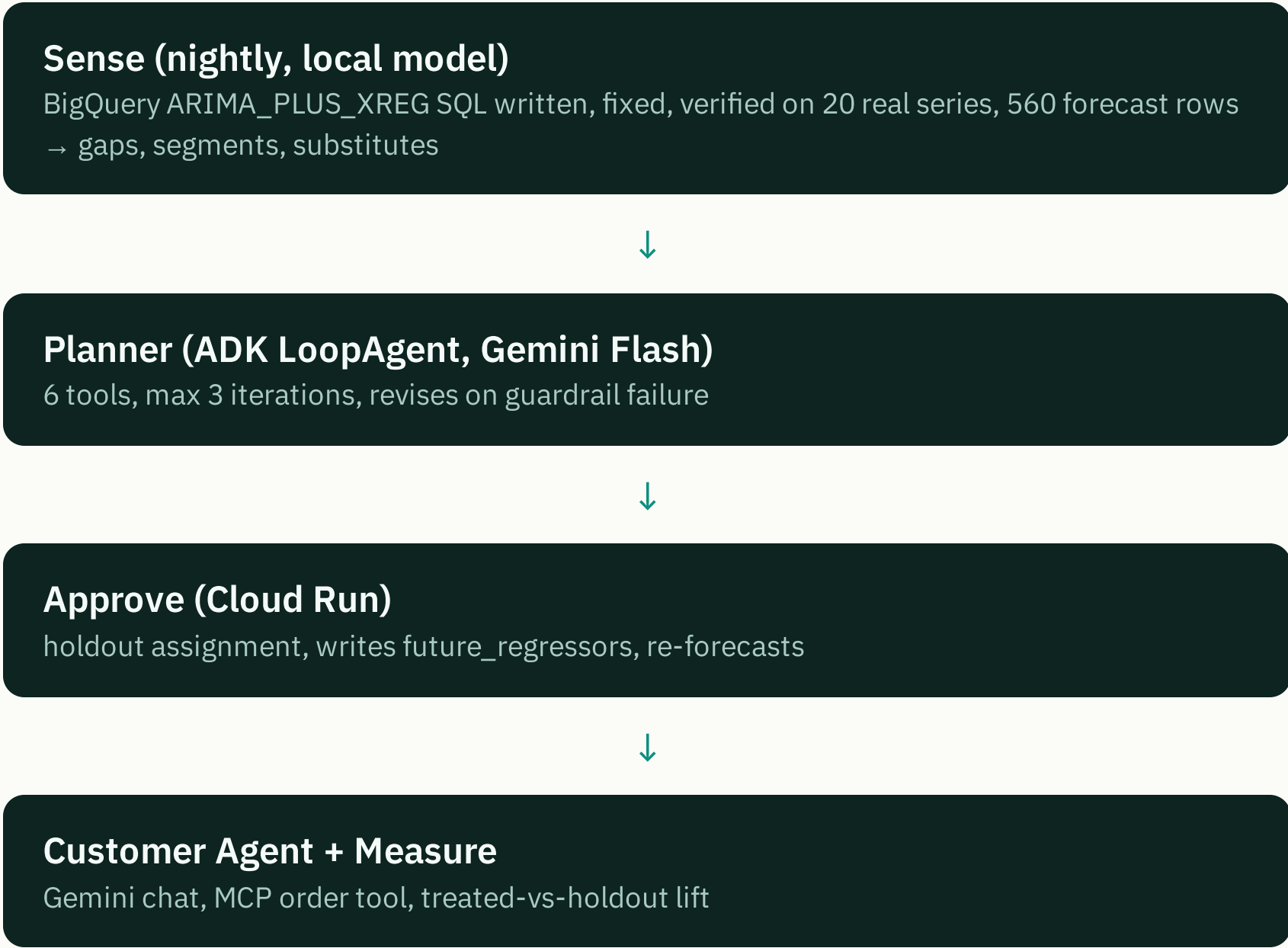
A decision layer: it emits an audience, copy and price a retailer's existing CRM or channel can send. It decides supply-side, per lot × node × deadline — not campaign scheduling.

Grounded, not general chat

The Customer Agent is grounded in this node's real stock and consent; it never promises a product the stock tool did not confirm.

TECHNICAL MERIT

Real services, measured live



Measured, live Vertex — not estimated

Path	Latency
Customer chat, p50	4.40 s
Customer chat, p95	5.75 s
Approve → re-forecast	8.30 s
Sense, 300 SKUs × 16 nodes	3.83 s

50 real chat calls against Gemini 2.5 Flash on Vertex, no caching, a fresh session per call. Full raw output committed to the repo, not typed by hand.

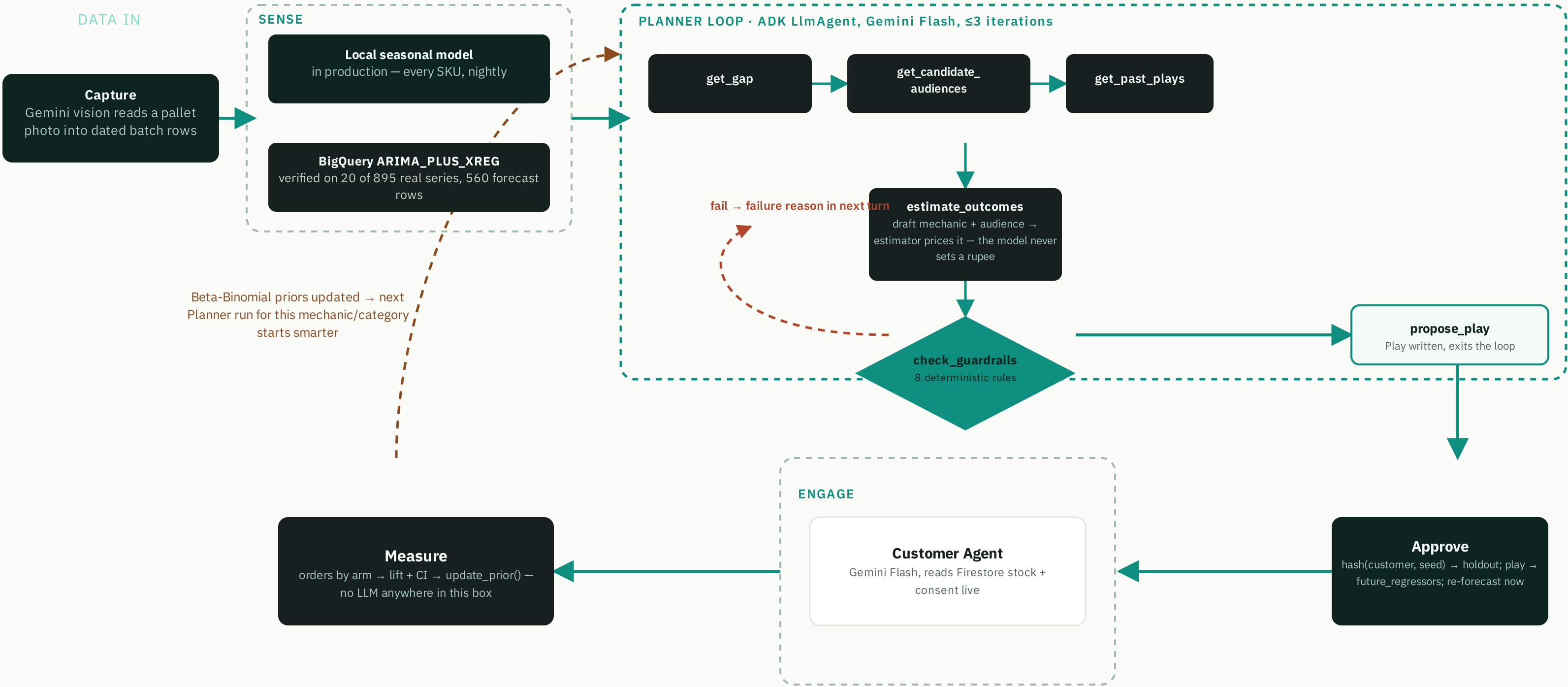
ARCHITECTURE

How data, agents and models fit together



ARCHITECTURE

System and agent orchestration, one lot's path through both



What Gemini decides — and what it is never allowed to

Gemini decides

- Which mechanic and audience for a gap, and the rationale
- Whether to revise a draft after a failed guardrail
- Reading a pallet photo into SKU, date, count, confidence
- Dialogue: substitution reasoning, the reply envelope
- Vernacular copy for approved plays (falls back to templates)

Deliberately never LLM

- Every rupee — the estimator, a pure function
- All 8 guardrails — margin floor, consent, holdout, ...
- Holdout assignment — a hash, reproducible from seed
- Stock, offer eligibility, order prices
- Lift, confidence intervals, measurement

The trace panel shows this live: a guardrail returning **passed: false**, and the exact next tool call the Planner makes to fix it — not a black box.

PROBLEM ALIGNMENT & IMPACT

Three ways to handle the same lot

368 units of Masala Chips, dark store DS-07, 6 days to online sell-by — the same lot from the opening slide

DO NOTHING ₹9,200 written off; estimator projection at plan time	BLANKET 20% MARKDOWN ₹8,189 discounted for everyone, including customers who'd have paid full price; estimator counterfactual	THIS PLAY, LIVE-APPROVED ₹9,194 → ₹8,068 targeted to 287 consented customers, real re-forecast, not simulated
Why now The FSSAI online sell-by rule already exists; quick-commerce and dark-store growth make the online window the binding constraint, not the shelf.	Consent-native, not a growth hack Every audience is filtered by marketing consent before a play exists; STOP withdraws it immediately, checked on every future play, not just at signup.	

Honest about the gap: this is the seeded demo tenant. No pilot with real customers has run yet — the Outcomes screen labels every row REAL PILOT or SYNTHETIC and shows **unmeasured** rather than guess a lift below the minimum sample size.

EVALUATION

Numbers we can defend

Metric	Result	Note
Agent Simulation, guardrail pass	190 / 190	live Vertex, 5 guardrails, 190 personas
Vision read accuracy, 30 photos	30 / 30	synthetic photos — not a real-camera test
Customer chat p50 / p95	4.40 s / 5.75 s	real gemini-2.5-flash, 50/50 live calls OK
Forecast backtest MAPE, 9 tiers	0.107–0.221	7 rolling origins, 63 rows
Sense throughput	3.83 s	300 SKUs × 16 nodes, one run
BigQuery ARIMA_PLUS_XREG, 20 of 895 series	560 rows	real job on a 324,271-row full load; 875 series not run
Pilot: treated vs holdout lift	not yet measured	no pilot has run
Cost per play, billing export	not yet measured	list-price estimate only, labelled

9 of 13 §12 rows carry a real, committed measurement; the rest say so rather than guess. Every number above is parsed from raw output under **eval/raw/**, never typed by hand.

LONGEVITY

Scalability and sustainability

Throughput, measured

300 SKUs × 16 nodes senses in 3.83 s. Cloud Tasks fan-out and per-gap triage by rupees-at-stake are the extrapolation path to a 20,000-SKU retailer.

Unit economics

Flash-only, no idle endpoints, nightly batch. A Cost Governor gates which gaps get a model call at all, by rupees at stake.

Onboarding

A three-CSV ingestion contract (products, nodes, batches) is the path from a seeded demo tenant to a real retailer's own catalogue.

Multi-tenant from day one

tenant_id on every table, per-tenant policy text — not a single-demo hack. Least-privilege IAM per service is written, not yet wired into the live deploy.

UX

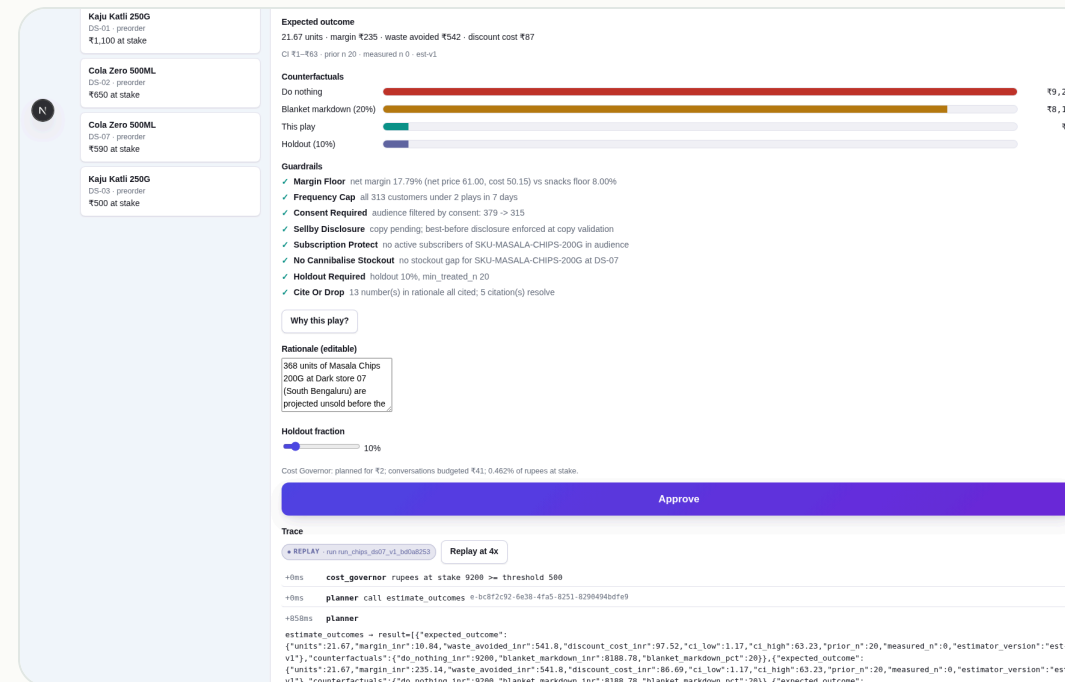
Three people, three screens, one job each

SKU GUESS	BEST BEFORE	FACINGS	CONFIDENCE
SKU-MASALA-CHIPS-200G	2026-11-02	2	sku 100% · date 100% · count 100%
SKU-SALTED-CHIPS-200G	2026-11-15	2	sku 80% · date 100% · count 100%
SKU-BANANA-CHIPS-200G	2026-10-28	2	sku 70% · date 100% · count 100%

Confirm rows

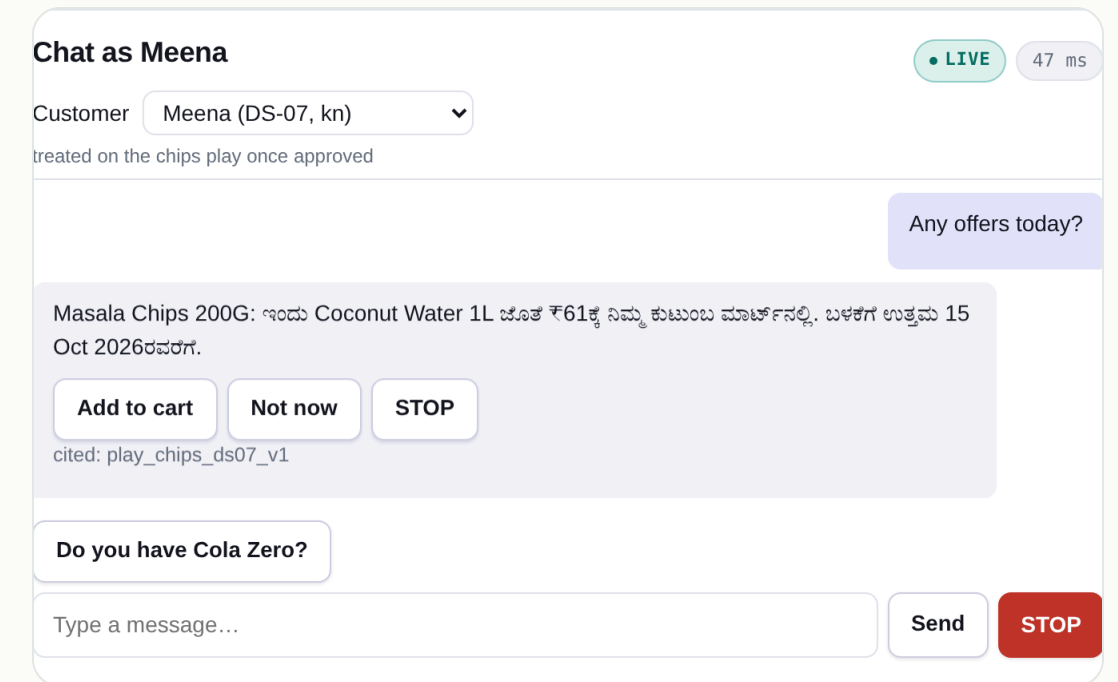
Priya, node manager

Photographs a pallet; low-confidence rows are confirmed, never assumed, before they reach inventory.



Arjun, demand planner

Reviews the rationale, the guardrails, the trace of a failed-then-revised draft, and approves.



Meena, customer

Gets the offer in her language with the best-before date, or an in-stock substitute, never a promise the stock tool didn't confirm.

Judge mode: no login, a per-visitor sandbox, "Run the 60-second beat" on the root URL, LIVE / REPLAY badges on every panel.

CLOSING

Every forecast tells a retailer what will be thrown away.
Taal is the agent that sells it first.

REAL

Forecasts (local model; BigQuery SQL written and verified on 20 of 895 series), gaps, estimator, guardrails, the Planner loop, holdout assignment, live re-forecast, chat, orders, measurement

SIMULATED, DISCLOSED

The tenant Kutumb Mart (seeded generator, its rule stated in code); no pilot has run yet — the Outcomes screen says so on every row